

2015 HKDSE Physics

Paper 1B Structured Questions

Original questions with official answers and English explanations. Correct answers are red; explanations are blue.

An asterisk marks extension content in the original paper.

Question 1

The solid curve in Figure 1.1 shows how the resistance of a metallic resistance thermometer varies with temperature. This thermometer is calibrated at standard atmospheric pressure for the melting point of ice and the steam point of boiling water. The dotted calibration line represents how the resistance of the thermometer varies with temperature if a linear resistance-temperature relationship is assumed. The deviation of the curve from linearity has been exaggerated in the figure.

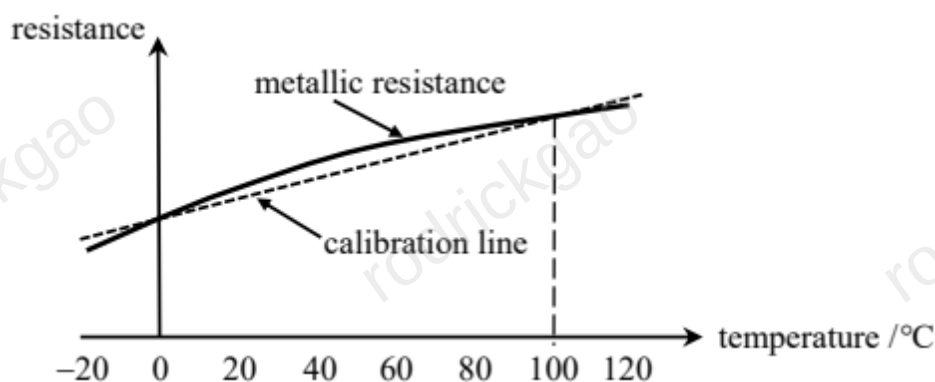


Figure 1.1

(a)(i) Using the resistances at the calibration points tabulated below, calculate the expected resistance at 60 °C if the resistance varies linearly with temperature. (2 marks)

Temperature / °C	Resistance / Ω
0	102.00
100	140.51

(a)(ii) Now if the resistance of the resistance thermometer is the value found in (a)(i), is the actual temperature higher than, lower than or equal to 60 °C? (1 mark)

(b) In an experiment to determine the specific heat capacity of water c_w , Peter used this calibrated resistance thermometer to measure the temperature of water being heated from 0 °C to 60 °C. Heating was stopped when this thermometer's resistance reached the value found in (a)(i). Assuming negligible heat exchange with the surroundings, no error in measuring the energy supplied and the mass of water, explain whether the experimental value of c_w found is higher than, lower than or the same as the actual value. (2 marks)

Official marking scheme

Solution		Marks	Remarks
1. (a)	(i) $\frac{140.51 - 102.00}{100 - 0} = \frac{R - 102.00}{60 - 0}$	1M	
	$R = 38.51 \times \frac{60}{100} + 102$	1A	
	$= 125 \, \Omega \, (125.106 \, \Omega)$	2	
	(ii) lower than 60 °C.	1A	
		1	
	(b) specific heat capacity = $\frac{\text{heat provided}}{\text{mass of water} \times \text{temp. rise}} \left[c = \frac{E}{m\Delta T} \right]$	1A	
	Since the actual final temperature is lower than 60 °C when heating is stopped, or actual $\Delta T < 60 \, ^\circ\text{C}$,	1A	
	Or heat provided is actually less than it should be when really reaching 60 °C	1A	
	the specific heat capacity is smaller than it should be.	1A	
		2	

(a)(i) Linear interpolation: $R = 102.00 + (60/100)(140.51 - 102.00) = 125.106 \, \Omega \approx 125 \, \Omega$. Use the two fixed-point resistances, not the curved graph.

(a)(ii) Lower than 60 °C. At a given temperature the solid curve lies above the calibration line, so the same resistance is reached at a lower actual temperature.

(b) The measured specific heat capacity is too low. The actual temperature rise is less than the assumed 60 °C. Since $c_w = E/(m\Delta T)$, using an overestimated ΔT with correct E and m underestimates c_w .

Question 2 *

The aqua-lung (a cylinder containing compressed air) for divers has a capacity of $1.0 \times 10^4 \text{ cm}^3$. When the aqua-lung is filled, the air inside has a pressure of 210 atm (atmospheric pressure) at 24°C . The air in the aqua-lung is allowed to expand through a pressure-reducing valve until its pressure equals that of the surrounding water before it is supplied to divers. Assume that the temperature of the air inside the aqua-lung is always equal to that of the surrounding water.

(a) A diver stays in water of temperature 24°C and pressure 2.0 atm at a depth of 10 m. Find the total volume of air (in cm^3) available for the diver from the aqua-lung at this water pressure. (2 marks)

(b) The supply of air in (a) is sufficient for the diver to remain at such a depth for 1 hour.

(b)(i) If the diver breathed in the same volume V_0 (in cm^3) of air per minute, find V_0 . (1 mark)

(b)(ii) If the diver dives deeper where the water is of temperature 20°C and pressure 4.5 atm, estimate how long (in minutes) the air in a fully-filled aqua-lung would last. Assume that the diver breathes in the same volume of air per minute as that found in (b)(i). (3 marks)

Official marking scheme

2. (a)	$210 \text{ atm} \times (1.0 \times 10^4 \text{ cm}^3) = 2.0 \text{ atm} \times V$ $V = 1.05 \times 10^6 \text{ cm}^3$ Volume available $= 1.05 \times 10^6 - 1.0 \times 10^4$ $= 1.04 \times 10^6 \text{ (cm}^3\text{)}$	1M	Accept ans. without considering residual volume, i.e. $1.05 \times 10^6 \text{ (cm}^3\text{)}$
		1A	
		2	
(b) (i)	$V_0 = 1.04 \times 10^6 \text{ cm}^3 \div 60$ $= 17333 \approx 17300 \text{ (cm}^3\text{) (per minute)}$	1M/1A	$V_0 = 17500 \text{ (cm}^3\text{) if residual volume not considered.}$
		1	
(ii)	V' : total volume of air at this depth/in this situation $\frac{P_1 V_1}{T_1} = \frac{P_2 V'}{T_2}$ $\frac{210 \times (1.0 \times 10^4)}{273 + 24} = \frac{4.5 \times V'}{273 + 20}$ $V' = 4.60 \times 10^5 \text{ cm}^3$ Volume available $= 4.60 \times 10^5 - 1.0 \times 10^4$ $= 4.50 \times 10^5 \text{ (cm}^3\text{)}$ Length of time : $= \frac{4.50 \times 10^5}{17333}$ $= 26.0 \text{ (min.)}$	1M	$V' = 4.60 \times 10^5 \text{ cm}^3$ and length of time $= 26.3 \text{ min. if residual volume not considered.}$
		1M	
		1A	
		3	

(a) At unchanged temperature, $p_1 V_1 = p_2 V_2$. All the air would occupy $V_2 = 210(1.0 \times 10^4)/2.0 = 1.05 \times 10^6 \text{ cm}^3$ at 2.0 atm. Subtract the $1.0 \times 10^4 \text{ cm}^3$ remaining in the cylinder: available volume $= 1.04 \times 10^6 \text{ cm}^3$.

(b)(i) $V_0 = (1.04 \times 10^6)/60 \approx 1.73 \times 10^4 \text{ cm}^3$ per minute.

(b)(ii) Use absolute temperatures: $T_1 = 297 \text{ K}$ and $T_2 = 293 \text{ K}$. From $pV/T = \text{constant}$, the full air supply occupies $V = 210(1.0 \times 10^4)(293)/(4.5 \times 297) \approx 4.60 \times 10^5 \text{ cm}^3$ at the new conditions.

(b)(ii) Available volume $= V - 1.0 \times 10^4 \approx 4.50 \times 10^5 \text{ cm}^3$. Duration $= (V - 1.0 \times 10^4)/(1.04 \times 10^6/60) \approx 26.0 \text{ min}$. Use the unrounded breathing rate.

The official scheme also accepts neglecting the residual cylinder volume: $1.05 \times 10^6 \text{ cm}^3$, $1.75 \times 10^4 \text{ cm}^3$ per minute, and approximately 26.3 min. Keep the chosen convention consistent.

Question 3

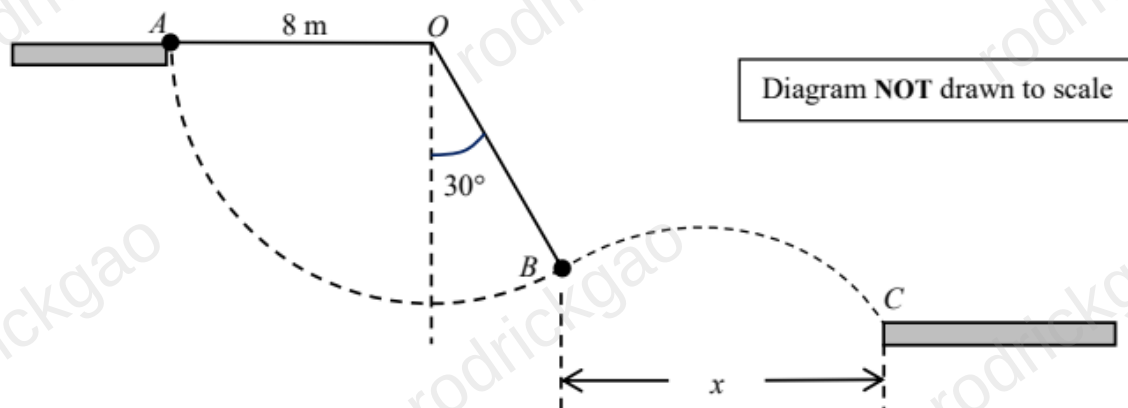


Figure 3.1

Figure 3.1 shows two horizontal platforms with end points A and C. An acrobat tries to swing from A to C by using a light rope of 8 m long and with one end fixed at point O, which is at the same level as A. He leaves A by holding the end of the rope and then releases it when reaching point B at which the angle between the rope and the vertical is 30° . The acrobat can be treated as a point mass and the rope remains taut and not extended throughout the motion. Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)

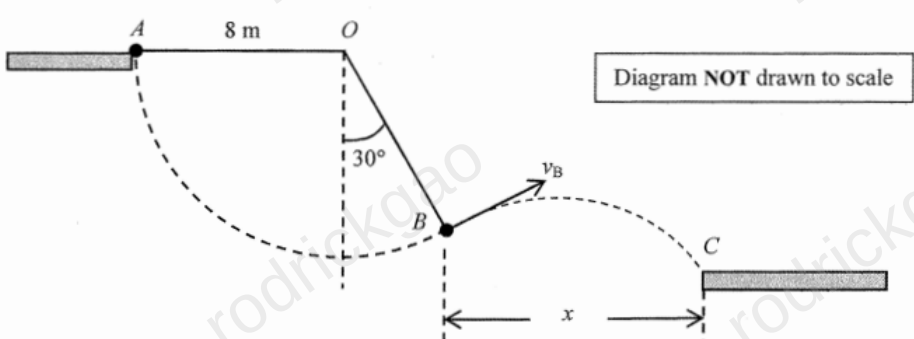
(a) Mark on Figure 3.1 the velocity v_B of the acrobat at B. If the speed of the acrobat when leaving A is zero, find the magnitude of v_B . (3 marks)

***(b)(i)** It takes 1.25 s for the acrobat to reach C after releasing the rope at B. By considering his horizontal motion, find the horizontal separation x between B and C. (2 marks)

***(b)(ii)** Calculate the vertical distance of C below B. (3 marks)

(c) Before reaching the lower platform, is there any change to the acrobat's mechanical energy among the points A, B and C? (1 mark)

Official marking scheme

Solution	Marks	Remarks
3. (a)  v_B correctly drawn with label (roughly perpendicular to OB). $\frac{1}{2}mv^2 = mgh$ $v_B^2 = 2gh = 2 \times 9.81 \times (8 \cos 30^\circ)$ $v_B = 11.7 \text{ m s}^{-1} \text{ (11.65896) (or } 11.77 \text{ m s}^{-1} \text{ for } g = 10 \text{ m s}^{-2})$ (b) (i) $x = v_x t = 11.7 \cos 30^\circ \times 1.25$ [$v_x = v_B \cos 30^\circ$] $= 12.6 \text{ m (12.62119)}$ (or 12.7 to 12.8 m for $g = 10 \text{ m s}^{-2}$) (ii) $y = ut - \frac{1}{2}gt^2$ $u = v_y = v_B \sin 30^\circ = 5.83 \text{ m s}^{-1}$ $y = v_y(1.25) - \frac{1}{2}(9.81)(1.25)^2$ $y = -0.38 \text{ m (-0.414 to -0.352 m)}$ (or -0.455 to -0.4375 m for $g = 10 \text{ m s}^{-2}$) Platform C is 0.38 m below B. (c) Total mechanical energy is the same / unchanged.	1A 1M 1A 3 1M 1A 2 1M 1M 1A 3 1A 1	

(a) Draw v_B tangent to the circular path at B, perpendicular to OB and directed upward to the right, at 30° above the horizontal. The fall in height is $8 \cos 30^\circ$. Conservation of energy gives $\frac{1}{2}mv_B^2 = mg(8 \cos 30^\circ)$, so $v_B = \sqrt{(2 \times 9.81 \times 8 \cos 30^\circ)} = 11.65896... \text{ m s}^{-1} \approx 11.7 \text{ m s}^{-1}$.

(b)(i) Horizontal velocity is constant after release: $x = (v_B \cos 30^\circ)(1.25) = 12.621... \text{ m} \approx 12.6 \text{ m}$.

(b)(ii) Take downward as positive. Initial vertical velocity is $-v_B \sin 30^\circ$. Thus the vertical distance below B is $y = -(v_B \sin 30^\circ)(1.25) + \frac{1}{2}(9.81)(1.25)^2 = 0.3772... \text{ m} \approx 0.38 \text{ m}$. Retain the unrounded v_B because the two terms nearly cancel.

(c) Mechanical energy is unchanged. Tension does no work because it is perpendicular to motion; after release only gravity does work, exchanging kinetic and gravitational potential energy.

Question 4

The motion of a block on an inclined plane can be investigated using a motion sensor connected to a computer (not shown in Figure 4.1).

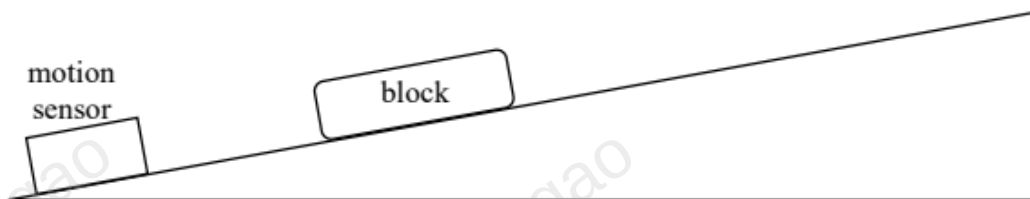


Figure 4.1

A block is given a push up a rough inclined plane and then released. The velocity-time (v - t) graph recorded by the sensor is shown below. Assume that the frictional force acting on the block is constant in magnitude throughout the motion. Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)

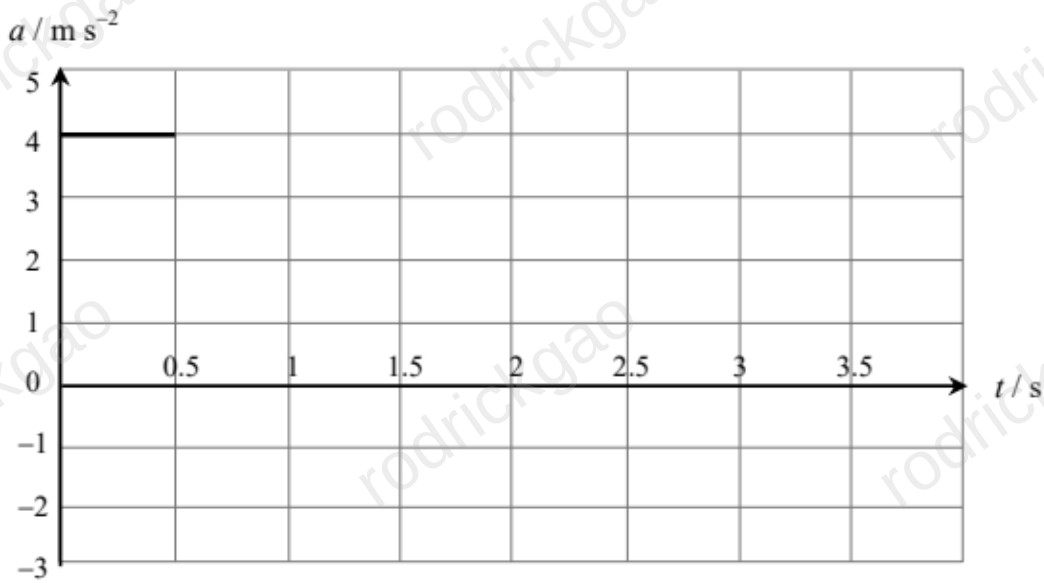


Point A on the graph corresponds to the instant at which the push is removed.

(a) Describe the block's motion from A to B. (2 marks)

(b)(i) Find the magnitude of the block's acceleration from $t = 1.5 \text{ s}$ to $t = 3.5 \text{ s}$. (2 marks)

(b)(ii) Draw the corresponding acceleration-time (a - t) graph of the block. With the direction up the inclined plane taken to be positive, the part during which the block is being pushed has been drawn for you. (2 marks)

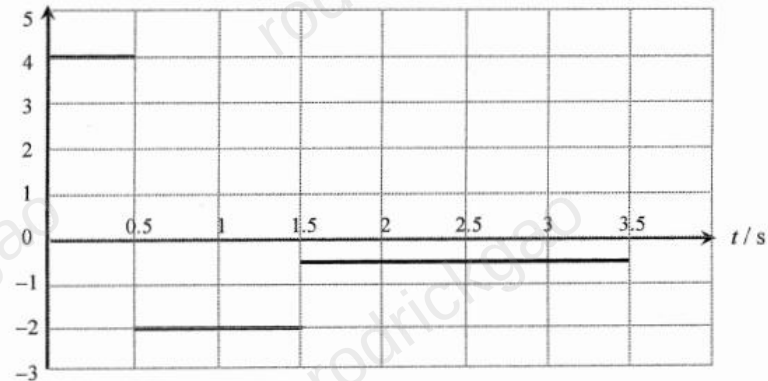
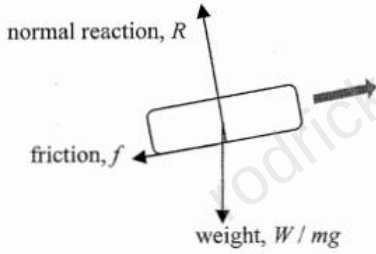


(c) Draw a free-body diagram to show the force(s) (with labels) acting on the block as it moves up the inclined plane after the push is removed. (2 marks)



(d) If the mass of the block is 1.0 kg, find the magnitude of the frictional force. (3 marks)

Official marking scheme

Solution		Marks	Remarks
4. (a)	The block decelerates uniformly (until at rest or its velocity is zero at $t = 1.5$ s). It then moves with uniform acceleration down the plane (until $t = 3.5$ s).	1A	
		1A	
		2	
(b) (i)	$a_2 = \frac{-1 - 0}{3.5 - 1.5}$ $= 0.5 \text{ m s}^{-2}$	1M	
		1A	
		2	
(ii)			
			
Either acceleration correct All correct		1A 1A 2	
(c)			
			
Friction correctly indicated All correct		1A 1A 2	
(d)			
Moving upward : $-mg \sin \theta - f = ma$ $-(1)(9.81) \sin \theta - f = (1)(-2) \dots \dots \dots \textcircled{1}$		1M	
Moving downward : $-mg \sin \theta + f = ma'$ $-(1)(9.81) \sin \theta + f = (1)(-0.5) \dots \dots \dots \textcircled{2}$		1M	
$\textcircled{2} - \textcircled{1} : 2f = 1.5$ $f = 0.75 \text{ N}$ (Note: $\theta = 7.32^\circ$)		1A	
		3	

(a) From 0.5 s to 1.5 s the block moves up the plane with uniform deceleration, coming momentarily to rest. It then reverses and moves down the plane with uniform acceleration until B.

(b)(i) Acceleration = $(-1 - 0)/(3.5 - 1.5) = -0.50 \text{ m s}^{-2}$; its magnitude is 0.50 m s^{-2} .

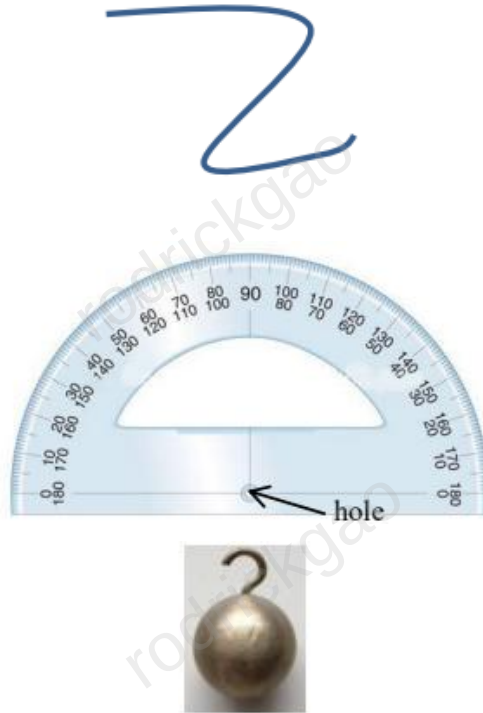
(b)(ii) The slope of the v-t graph gives $a = +4 \text{ m s}^{-2}$ for $0 < t < 0.5 \text{ s}$ (already drawn), -2 m s^{-2} for $0.5 < t < 1.5 \text{ s}$, and -0.5 m s^{-2} for $1.5 < t < 3.5 \text{ s}$. Draw horizontal segments with jumps at the changes.

(c) Draw three forces only: weight mg vertically downward; normal reaction perpendicular to and away from the plane; friction down the plane. The push has been removed.

(d) Let the plane's angle be θ and friction magnitude be f . During upward motion, $mg \sin \theta + f = 2m$. During downward motion, $mg \sin \theta - f = 0.5m$. Subtracting gives $2f = 1.5m$, so $f = 0.75 \text{ N}$ for $m = 1.0 \text{ kg}$.

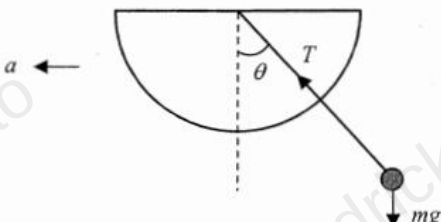
Question 5

You are given a long light string, a protractor and a metal ball with a hook.



Suppose you are inside a train which is at rest initially and later it travels along a straight horizontal track with constant acceleration. With the aid of a diagram, describe how to measure the acceleration of the train. Show your working including mathematical derivation. (6 marks)

Official marking scheme

Solution	Marks	Remarks
5. Diagram  Tie one end of the string to the metal ball and the other end through the centre/hole of the protractor. When the train is at rest, held fixed the protractor in the plane along the direction of motion such that the string is on, say, the 90° mark. When the train is accelerating with acceleration a, the string will make an angle, say θ, with the vertical. Measure the angle θ. Let T be the tension of the string, Vertically : $T \cos \theta = mg$ ① Horizontally : $T \sin \theta = ma$ ② where m is the mass of the ball ② : $\tan \theta = \frac{a}{g}$ ① : $a = g \tan \theta$	1A 1A 1A 1M 1A 6	

Set-up: Tie the metal ball to the string and suspend it through the hole at the protractor's centre. Hold the protractor fixed inside the train with its plane parallel to the direction of motion. The official diagram above shows the arrangement and forces.

Reference reading: With the train at rest, record the direction of the freely hanging string; this is the vertical reference.

Measurement: During constant acceleration, wait until the ball settles at a steady backward tilt. Read the angle θ between the string and the vertical. Keep the support fixed and avoid taking a reading while the ball swings.

Derivation: The ball has the same horizontal acceleration a as the train and no vertical acceleration. Resolving tension T gives $T \cos \theta = mg$ and $T \sin \theta = ma$.

Result: Dividing gives $\tan \theta = a/g$, hence $a = g \tan \theta$. Use $g = 9.81 \text{ m s}^{-2}$ and the measured θ .

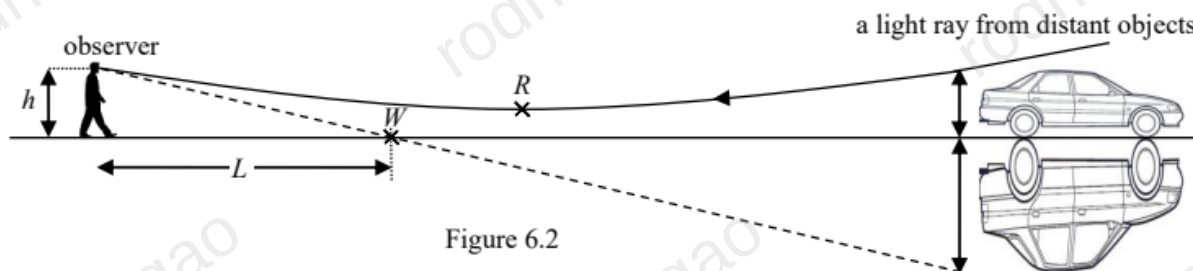
Question 6

Read the following description about a mirage and answer the questions that follow.

A mirage is often seen on highways during hot summers. Pools of water seem to cover the roadway far ahead. Distant objects appear to be reflected by the surface of the “water”. The phenomenon is caused by the difference in refractive index between the hot air near the road surface and the cooler air above it. The refractive index of cool air is greater than that of hot air, but the differences are so small that the subsequent deviations of light rays are tiny. Sufficiently large temperature differences between the hot air near the road surface and the above cooler air over a short height (i.e. high temperature gradient) and light rays travelling along sufficiently long path lengths are required to form a mirage.



Figure 6.1 Mirage seen on a highway. This photo was taken with a telephoto lens which gives the perception that the viewer is very close to the car ahead.



Figures 6.2 and 6.3 illustrate the principle of the phenomenon. Air of different temperatures is simplified to several layers and modeled as parallel slabs as shown in Figure 6.3. The bending of the light ray from distant objects is much exaggerated. θ_1 , θ_2 , θ_3 and θ_4 denote the angles of incidence at various boundaries of air layers.

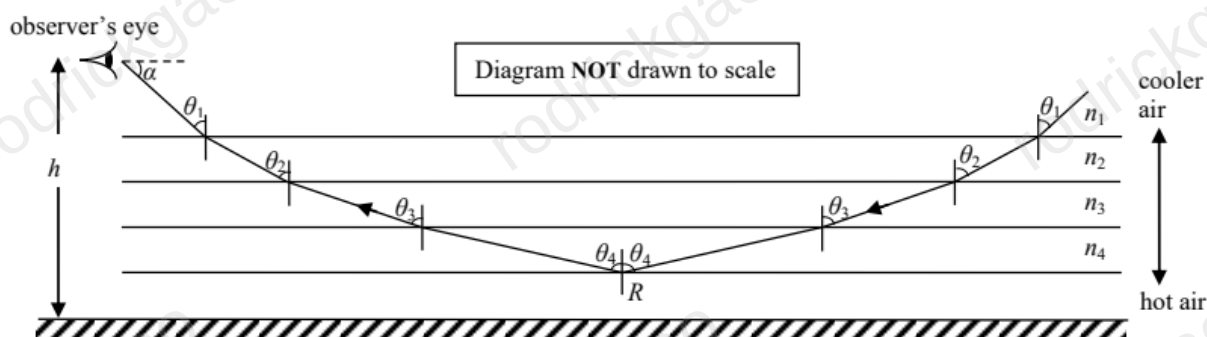


Figure 6.3

- (a) State ONE essential condition for a mirage to be observed. (1 mark)
- (b)(i) Referring to Figure 6.3, deduce the relationship between θ_1 , θ_4 and refractive indices n_1 , n_4 . For total internal reflection just to occur at R , θ_4 can be taken as 90° . Hence, find the corresponding value of θ_1 if $n_1 = 1.000261$ and $n_4 = 1.000221$. (3 marks)
- (b)(ii) Find L in Figure 6.2 if $h = 1.5$ m. (Note: $\alpha + \theta_1 = 90^\circ$ in Figure 6.3.) (2 marks)
- (c) A thirsty traveller in a vast desert sees similar mirages such that a “water source” appears at W which is distance L away like the one in Figure 6.2. If he walks a distance L towards that “water source”, how far would the “water source” appear to him? Explain your answer. (2 marks)

Official marking scheme

Solution	Marks	Remarks
6. (a) - high temperature gradient; <u>or</u> - long path lengths for light rays.	1A	Accept 167.7 m to 172.0 m
<u>Or</u> total internal reflection occurs.	1A	
	1	
(b) (i) $n_1 \sin \theta_1 = n_2 \sin \theta_2 = n_3 \sin \theta_3 = n_4 \sin \theta_4$	1M	
$\sin \theta_1 = \frac{n_4}{n_1} \sin \theta_4$		
$\theta_1 = \sin^{-1} \left(\frac{1.000221}{1.000261} \right)$	1M	
$= 89.5^\circ (89.488^\circ)$	1A	
	3	
(ii) $\frac{h}{L} = \tan \alpha = \frac{1}{\tan \theta_1}$	1M	
$L = h \tan \theta_1 = 1.5 \tan 89.5^\circ = 167.72$		
$= 168 \text{ m}$	1A	
<u>Or</u> $L = \frac{h}{\tan \alpha} = \frac{1.5}{\tan 0.5^\circ} = 171.88$	1M	
$= 172 \text{ m}$	1A	
	2	
(c) The same distance away (168 m) because the illusion of 'water source' is caused by reflection of the light of distant objects at the same fixed angle. [i.e. $\alpha = 90^\circ - 89.488^\circ = 0.512^\circ$ with the horizontal]	1A 1A	
<u>Or</u> As long as the conditions for bending of the light and internal reflection are still the same, the 'water source' remains 168 m away (satisfying the same conditions / angle of reflection).	1A 1A	
	2	

(a) One essential condition is a sufficiently large vertical temperature gradient: air near the road is much hotter than the air just above. The official scheme also accepts a sufficiently long light path or the condition for total internal reflection.

(b)(i) Apply Snell's law successively at the parallel boundaries: $n_1 \sin \theta_1 = n_2 \sin \theta_2 = n_3 \sin \theta_3 = n_4 \sin \theta_4$. Hence $n_1 \sin \theta_1 = n_4 \sin \theta_4$.

(b)(i) At the limiting condition $\theta_4 = 90^\circ$, $\sin \theta_1 = 1.000221/1.000261$. Therefore $\theta_1 = 89.4876...^\circ \approx 89.5^\circ$.

(b)(ii) Since $\alpha = 90^\circ - \theta_1$, $L = h/\tan \alpha = h \tan \theta_1$. Using the unrounded angle gives $L = 167.7... \text{ m} \approx 168 \text{ m}$. Substituting 89.5° prematurely changes this sensitive result.

(c) It would still appear about L (168 m) away, provided the same atmospheric conditions and eye height apply. The required ray angles, and hence $h/\tan \alpha$, remain the same at the new position; W is an apparent image, not a fixed pool of water.

Question 7

*(a) A laser light beam of wavelength 650 nm is incident normally on a pair of slits separated by $a = 0.325$ mm. Interference pattern is observed on a screen at a distance $D = 3.0$ m from the slits as shown in Figure 7.1. What is the separation between adjacent first- and second-order bright fringes? (2 marks)

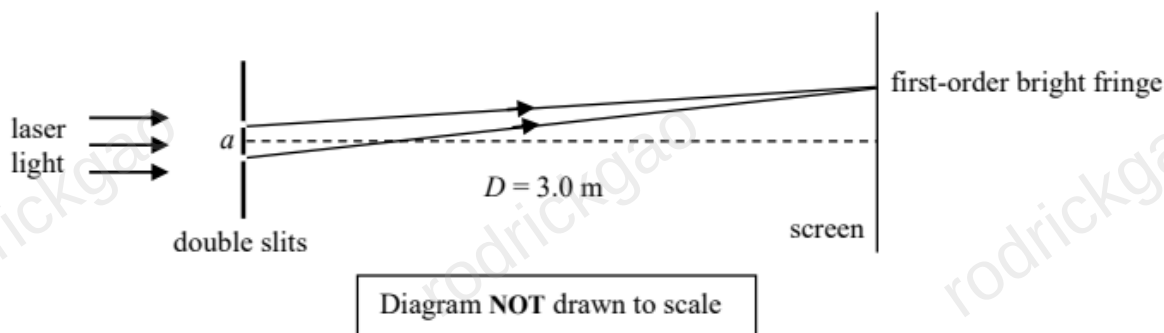


Figure 7.1

(b) Figure 7.2 shows a set-up with two small LEDs separated by 6 mm and both LEDs emit light of wavelength 650 nm. State and explain what you would expect to see on the screen. (2 marks)

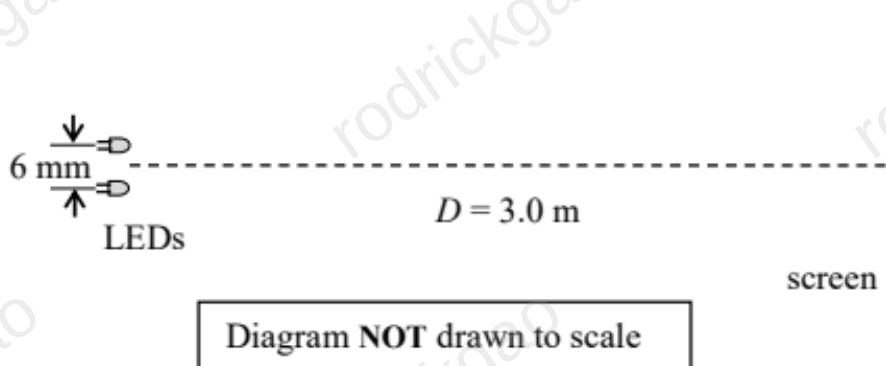


Figure 7.2

Figure 7.3 shows circular water waves in a ripple tank. The two point sources S_1 and S_2 , separated by 40 mm, are driven by the same vibrator. The solid lines represent the wave crests from S_1 and the dotted lines represent the wave crests from S_2 . The wavelength of the waves is 10 mm.

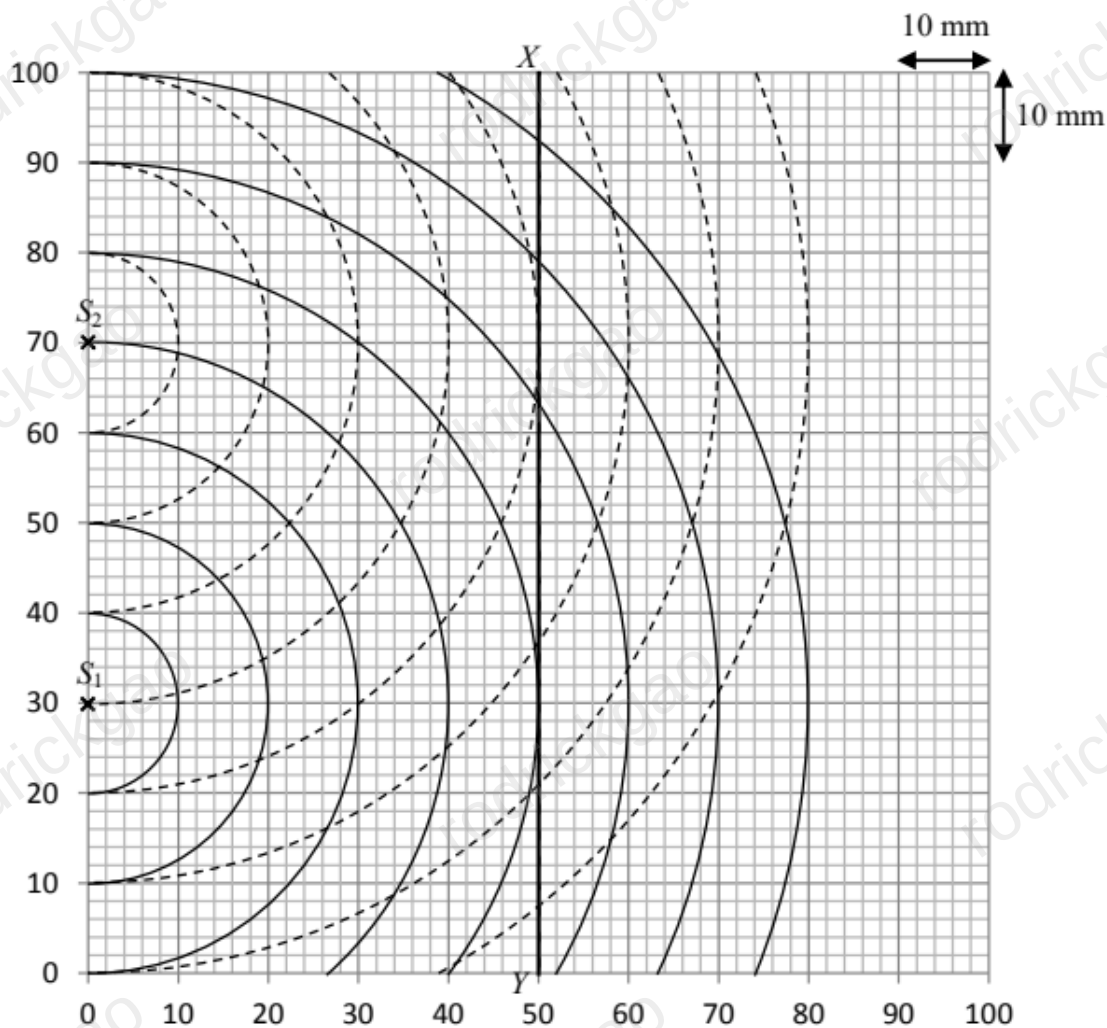


Figure 7.3

(c) Sketch on Figure 7.3 two lines to indicate all points P with path difference $PS_1 - PS_2$ equal to 10 mm (L_1) and 20 mm (L_2). State the kind of interference that occurs at these points P. (3 marks)

(d)(i) If the interference pattern is observed along line XY at 50 mm from the sources as shown, measure the separation between adjacent first- and second-order maxima Δy . (1 mark)

*(d)(ii) However, using the calculation method in (a) would obtain 12.5 mm for this separation. Why does this calculated value differ with the measurement in (d)(i)? (2 marks)

Official marking scheme

Solution	Marks	Remarks
7. (a) $\Delta y = \frac{\lambda D}{a}$ $= \frac{(650 \times 10^{-9}) \times 3.0}{0.325 \times 10^{-3}}$ $= 0.006 \text{ m or } 6 \text{ mm}$ (b) The screen is uniformly illuminated (The interference patterns exist very briefly and change rapidly such that, to human eyes, they are averaged out). The light from the LEDs is incoherent (i.e. no fixed phase relationship between the light coming out from the two LEDs). (c) path difference $PS_1 - PS_2 = 10 \text{ mm}$, L_1 correct. path difference $PS_1 - PS_2 = 20 \text{ mm}$, L_2 correct. Constructive interference (occurs at P)	1M 1A 2 1A 1A 2 1A 1A 1A 3	NOT accept 'interference does not occur'.
(d) (i) $\Delta y = y_2 - y_1 = 31 \text{ mm} - 14 \text{ mm} = 17 \text{ mm} \pm 2 \text{ mm}$ (ii) Screen has to be far away from slits or $D \gg a$, (i.e. to satisfy $D \gg y$ / consider y to be close to the central maximum) Or screen is too close to slits or $D \gg a$ not satisfied (i.e. $D \gg y$ not satisfied) Make use of small angle approximation ($\theta \approx \sin \theta \approx \tan \theta$ / small angle approximation cannot be applied.	1A 1 1A 1A 1A 2	

- (a) Adjacent bright-fringe separation is $\Delta y = \lambda D/a = (650 \times 10^{-9})(3.0)/(0.325 \times 10^{-3}) = 6.0 \times 10^{-3} \text{ m} = 6.0 \text{ mm}$.
- (b) The screen is illuminated without a stable bright-and-dark interference pattern. The two independent LEDs are not coherent: their phase difference varies randomly. Equal wavelength alone is insufficient.
- (c) Positive $PS_1 - PS_2$ selects the side nearer S_2 (the upper half of the supplied grid). Draw smooth loci through crest intersections having path differences λ and 2λ ; these pass through the vertical source line at grid heights 55 mm and 60 mm respectively. The official diagram shows the two branches. Both are constructive-interference maxima.
- (d)(i) Read the intersections of L_1 and L_2 with XY using the printed grid scale. Their separation is about 17 mm; the official tolerance is $\pm 2 \text{ mm}$. Do not measure the resized picture in physical millimetres.
- (d)(ii) The formula $\Delta y = \lambda D/a$ assumes a distant screen and small angles (D much greater than a , and y much smaller than D). Here $D = 50 \text{ mm}$ is comparable to $a = 40 \text{ mm}$, so the approximation fails and the maxima are not uniformly spaced.

Question 8

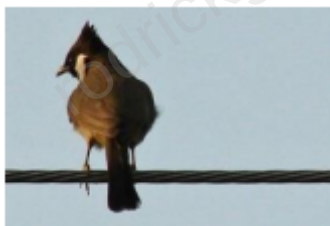
Electricity generated from power plants are transmitted at a high voltage through overhead cables in suburban areas.

- (a) Each overhead cable consists of 40 strands of identical transmission lines bundled together.



a strand of transmission line of an overhead cable

- (a)(i) One single strand of transmission line has a cross-sectional area of $1.3 \times 10^{-5} \text{ m}^2$ and resistivity $2.6 \times 10^{-8} \Omega \text{ m}$. Find the resistance per km of a single strand of transmission line. (2 marks)
- (a)(ii) Explain why the resistance per km of an overhead cable is much smaller than that of a single strand of transmission line. Estimate the resistance per km of an overhead cable. (2 marks)
- (a)(iii) Hence, explain why a bird can stand with both feet on a high-voltage cable without getting an electric shock. (2 marks)



- * (b) Electrical power of 180 MW is transmitted at a voltage of 400 kV through an overhead cable.
- * (b)(i) Calculate the current carried by the overhead cable. (2 marks)
- * (b)(ii) Show that less than 0.1% of the electrical power is lost after transmitted through a total of 10 km of overhead cable. (2 marks)
- * (b)(iii) As the voltage drop across this overhead cable is negligible, a voltage of 400 kV at the cable's end is stepped down by an ideal transformer with turns ratio 12:1.
- * (b)(iii)(I) Find the secondary voltage from the transformer. (1 mark)
- * (b)(iii)(II) State ONE factor leading to energy loss in a practical transformer and suggest the corresponding measure for improvement. (2 marks)

Official marking scheme

Solution	Marks	Remarks
8. (a) (i) $\rho = \frac{RA}{l}$ $\frac{R}{l} = \frac{\rho}{A} = \frac{2.6 \times 10^{-8}}{1.3 \times 10^{-5}}$ $= 2.0 \times 10^{-3} \Omega \text{ m}^{-1}$ $= 2.0 \Omega \text{ km}^{-1} \text{ or } 2.0 \Omega$	1M 1A 2	
(ii) The strands of transmission lines are in parallel / The cross-sectional area of cable is larger than that of each transmission line / Resistance is inversely proportional to the cross-sectional area of the cable $R_{\text{cable}} = \frac{R}{40} = 0.05 \Omega \text{ km}^{-1} \text{ or } 0.05 \Omega$ $\left(\frac{1}{R_{\text{cable}}} = \frac{1}{R} + \frac{1}{R} + \dots + \frac{1}{R} \rightarrow \frac{1}{R_{\text{cable}}} = \frac{40}{R} \right)$	1A 1M 2	
(iii) The resistance of the bird's body is much larger than that of a short segment of the overhead cable. Or The bird is in parallel with a short segment of the overhead cable. The potential difference across its feet is very small (very small resistance per km). Hence, negligible current flows through the bird's body.	1A 1A 2	
(b) (i) $I = \frac{P}{V} = \frac{180 \times 10^6}{400 \times 10^3}$ $= 450 \text{ A}$	1M 1A 2	
(ii) Percentage of power loss = $\frac{P_{\text{loss}}}{P_{\text{total}}} \times 100\%$ $= \frac{450^2 \times 0.05 \times 10}{180 \times 10^6} \times 100\%$ $= 0.05625 \% < 0.1 \%$	1M 1A 2	
(iii) (I) $N_p : N_s = V_p : V_s$ $12 : 1 = 400 : V_s$ $V_s = 33.3 \text{ kV}$	1A 1	
(II) Any ONE of the followings: Resistance of coils + use thicker wire for the coils / Magnetisation and demagnetisation of core + use soft iron core / Induced eddy currents in core + laminated core / Flux leakage + core design	1A+1A 2	

(a)(i) $R = \rho l/A = (2.6 \times 10^{-8})(1000)/(1.3 \times 10^{-5}) = 2.0 \, \Omega$ for 1 km.

(a)(ii) The 40 equal strands are in parallel, so the cable resistance is $2.0/40 = 0.050 \, \Omega$ per km. Equivalently, its conducting cross-sectional area is 40 times larger.

(a)(iii) The short cable length between the bird's feet has very small resistance, so the potential difference between its feet is very small. Only negligible current passes through the bird. This applies to both feet on the same cable, without contact with another potential.

(b)(i) $I = P/V = (180 \times 10^6)/(400 \times 10^3) = 450 \, \text{A}$.

(b)(ii) Total cable resistance = $10(0.050) = 0.50 \, \Omega$. Power loss = $I^2 R = 450^2(0.50) = 101250 \, \text{W}$. Percentage loss = $[101250/(180 \times 10^6)] \times 100\% = 0.05625\%$, below 0.1%.

(b)(iii)(I) $V_s/V_p = N_s/N_p = 1/12$, so $V_s = 400/12 = 33.3 \, \text{kV}$.

(b)(iii)(II) One matched pair: eddy currents cause heating in the iron core; use a laminated core with electrically insulated sheets to reduce those currents.

Question 9

Figure 9.1 shows a set-up for demonstrating one of Faraday's discoveries. A light metal rod is free to rotate about point P while its lower end just touches some conducting liquid in a metallic container.

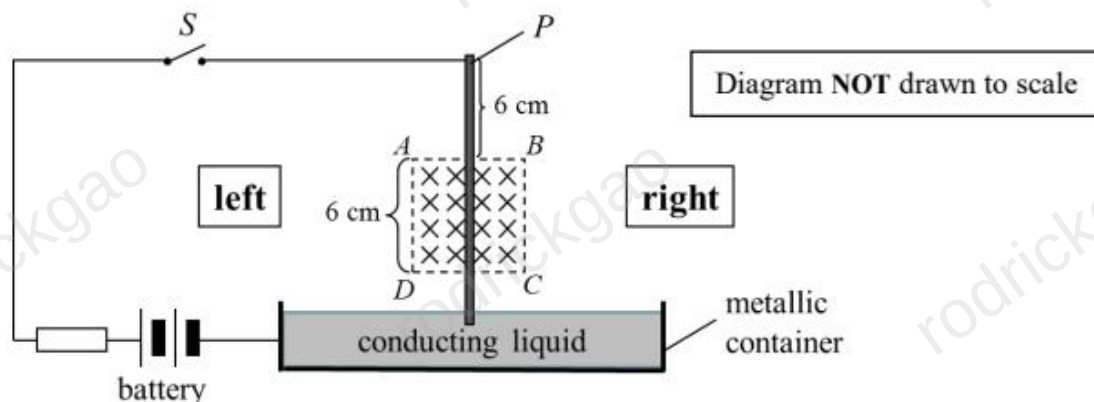


Figure 9.1

A uniform magnetic field pointing into the paper is applied over the region ABCD containing part of the rod. When switch S is closed, the rod “kicks” out and leaves the liquid surface.

- (a) State the direction (to the left / to the right / into the paper / out of the paper) that the rod “kicks” and describe the subsequent motion of the rod. (3 marks)
- (b) When switch S is closed, the initial moment about point P that makes the rod “kick” out is $7.2 \times 10^{-4} \text{ N m}$. Assume that the magnetic force acts at the midpoint of the part of the rod within the magnetic field.
 - (b)(i) Calculate the magnetic force acting on the rod at that instant. (2 marks)
 - (b)(ii) Hence, find the strength B of the magnetic field if the current flowing through the rod is 3.2 A when the circuit is closed. (2 marks)
- (c) Now the uniform magnetic field is removed and a bar magnet is placed underneath the container as shown in Figure 9.2. The rod is held tilted at an angle to the vertical but with its lower end still in the conducting liquid.

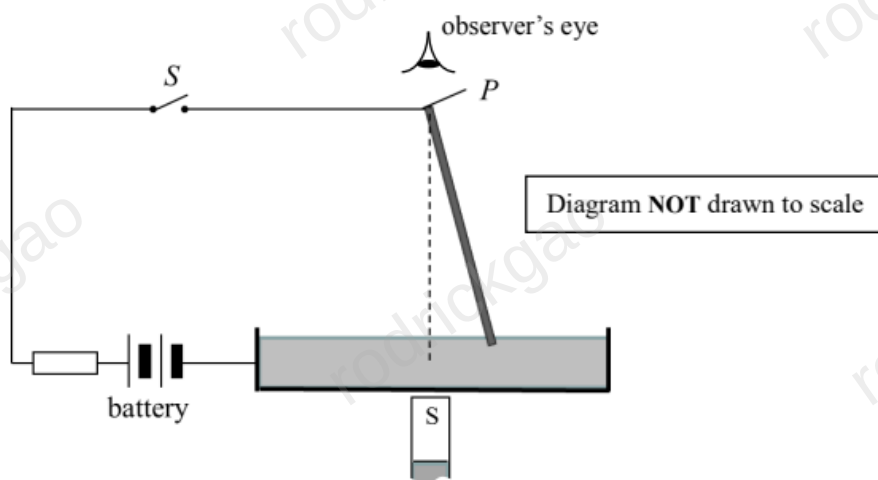
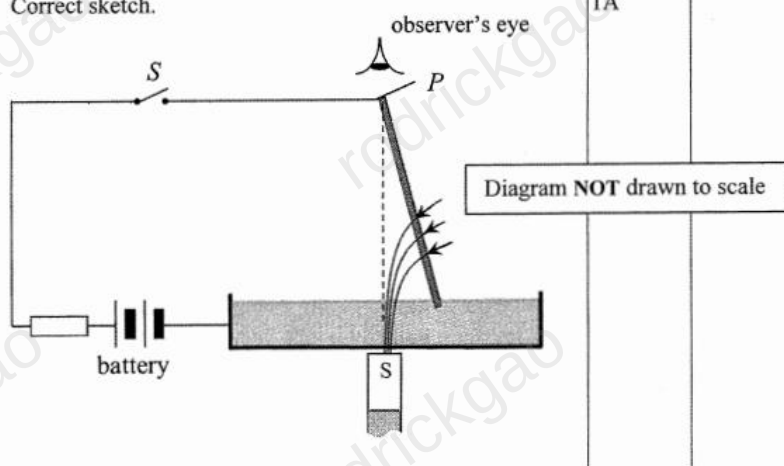


Figure 9.2

- (c)(i) Sketch on Figure 9.2 the field lines around the rod due to the bar magnet. (1 mark)
- (c)(ii) After closing switch S and the rod is released from rest, describe its subsequent motion viewed from above. (1 mark)

Official marking scheme

Solution	Marks	Remarks
9. (a) Right (current flowing downward, B -field into paper) When the rod reaches the highest point it falls. Then its lower end touches the conducting liquid again and the same magnetic force makes it 'kick' out from the liquid. The process repeats so the rod continually 'kicks' out and then returns.	1A 1A 1A	
	3	
(b) (i) As moment = $F \times d$ $7.2 \times 10^{-4} \text{ N m} = F (0.09 \text{ m})$ $F = \frac{7.2 \times 10^{-4}}{0.09} = 8.0 \times 10^{-3} \text{ N}$	1M 1A	
	2	
(ii) $F = BIl$ $8.0 \times 10^{-3} \text{ N} = B (3.2 \text{ A}) (0.06 \text{ m})$ $B = 0.042 \text{ T}$	1M 1A	
	2	
(c) (i) Correct sketch.	1A	
		
	1	
(ii) The rod will rotate anti-clockwise (as viewed from above).	1A	
Or The rod describes a conical pendulum.	1A	
	1	

(a) The rod kicks to the right. Conventional current is downward and the field is into the paper, giving a rightward magnetic force. Once the rod leaves the liquid, the circuit opens and the magnetic force ceases. It falls back, reconnects and kicks again, repeating the motion.

(b)(i) The perpendicular lever arm is $6 \text{ cm} + \frac{1}{2}(6 \text{ cm}) = 9 \text{ cm} = 0.090 \text{ m}$. Thus $F = \text{moment}/d = (7.2 \times 10^{-4})/0.090 = 8.0 \times 10^{-3} \text{ N}$.

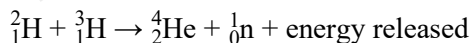
(b)(ii) Only 0.060 m of the rod is in the uniform field. From $F = BIl$, $B = (8.0 \times 10^{-3})/(3.2 \times 0.060) = 0.0417 \text{ T} \approx 0.042 \text{ T}$. Do not use the 0.090 m lever arm as l .

(c)(i) Draw field lines directed into the south pole of the bar magnet; the official sketch shows their curvature around the rod.

(c)(ii) The rod rotates anticlockwise when viewed from above. The current in the rod and the inward component of the magnetic field produce a tangential magnetic force.

Question 10

Scientists had been experimenting controlled fusion in a nuclear reactor in which deuterium (${}^2_1\text{H}$) and tritium (${}^3_1\text{H}$) undergo the following nuclear fusion:



Given: mass of a deuterium nucleus = 2.014102 u

mass of a tritium nucleus = 3.016049 u

mass of a helium nucleus = 4.002602 u

mass of a neutron = 1.008665 u

***(a)** Calculate the energy released, in MeV, in the above nuclear fusion. (2 marks)

(b) A deuterium nucleus and a tritium nucleus have to be within 10^{-15} m for nuclear fusion to occur and a large amount of work done (about 0.4 MeV) is needed to bring two well separated nuclei to such a close distance.

(b)(i) Explain why a large amount of work done is needed and state the kind of energy this work done has become. (2 marks)

In the nuclear reactor, deuterium and tritium exist as plasma, which is a mixture of ions at a very high temperature.

(b)(ii) Explain why a very high temperature is needed for nuclear fusion to occur. (1 mark)

***(b)(iii)** Estimate the order of magnitude of the minimum temperature at which fusion of deuterium and tritium nuclei would be possible if the plasma can be regarded as an ideal gas. (Hint: For an ideal gas, each gas molecule is assumed to have an average kinetic energy $E_K = 3RT/(2N_A)$.) (2 marks)

Official marking scheme

Solution	Marks	Remarks
10. (a) Mass deficit $= (2.014102 + 3.016049) \text{ u} - (4.002602 + 1.008665) \text{ u}$ $= 0.018884 \text{ u}$ Energy released $= 0.018884 \times 931 \text{ MeV}$ $= 17.58 \text{ (MeV)}$	1M 1A	
Or Energy released $= 0.018884 \times 1.661 \times 10^{-27} \times c^2$ $= 2.823 \times 10^{-12} \text{ J or } 17.64 \text{ MeV}$	1A	
	2	
(b) (i) To overcome the (electrostatic) repulsion between the two (positive) nuclei and becomes electrical potential energy (of the two nuclei).	1A 1A	
	2	
(ii) High temperature enables them to have sufficient kinetic energy (to overcome electrical repulsion between the nuclei).	1A	
	1	
(iii) Kinetic energy becomes electrical potential energy $E_p = 2 \times \frac{1}{2} m (c_{\text{rms}})^2 = 2 \times \frac{3RT}{2N_A}$ $0.4 \text{ MeV} = 2 \times \left(\frac{3 \times 8.31 \times T}{2 \times 6.02 \times 10^{23}} \right)$ $T = 1.545 \times 10^9 \text{ K i.e. order of magnitude } 10^9 \text{ (K)}$	1M 1A	Accept without the factor '×2' $0.4 \text{ MeV} = 6.4 \times 10^{-14} \text{ J}$
Alternatively: $E_p = \frac{1}{4\pi\epsilon_0} \cdot \frac{e^2}{10^{-15}} = 2 \times \frac{3RT}{2N_A}$ $T = 5.56 \times 10^9 \text{ K i.e. order of magnitude } 10^9 \text{ (K)}$	1M 1A	
	2	

(a) Mass defect $\Delta m = (2.014102 + 3.016049) - (4.002602 + 1.008665) = 0.018884 \text{ u}$. Using $1 \text{ u} = 931 \text{ MeV}/c^2$, energy released $= 0.018884 \times 931 = 17.58 \text{ MeV} \approx 17.6 \text{ MeV}$.

(b)(i) Both nuclei are positively charged, so bringing them together requires work against electrostatic repulsion. This work increases their electric potential energy.

(b)(ii) A high temperature provides sufficiently large kinetic energies for the nuclei to approach closely enough for fusion.

(b)(iii) For the two colliding nuclei, estimate the available total kinetic energy as $2 \times 3RT/(2N_A)$. Set this equal to $0.4 \text{ MeV} = 6.4 \times 10^{-14} \text{ J}$.

(b)(iii) $T \approx (6.4 \times 10^{-14})(6.02 \times 10^{23})/(3 \times 8.31) = 1.55 \times 10^9 \text{ K}$. The required order of magnitude is 10^9 K . The official scheme also accepts a one-particle estimate giving the same order.